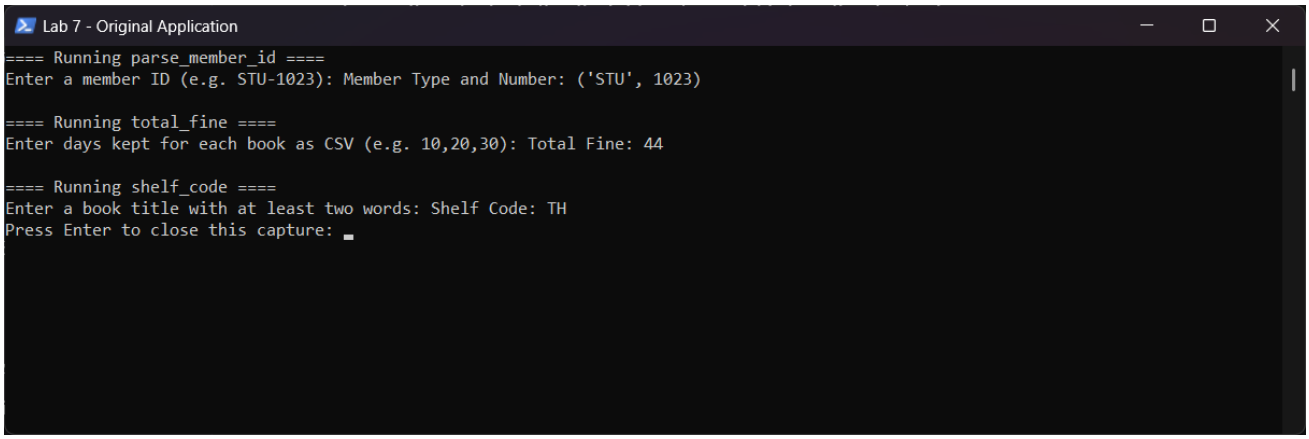
LAB 7 - FUZZ TESTING WITH HYPOTHESIS

Library Fines Input Validation

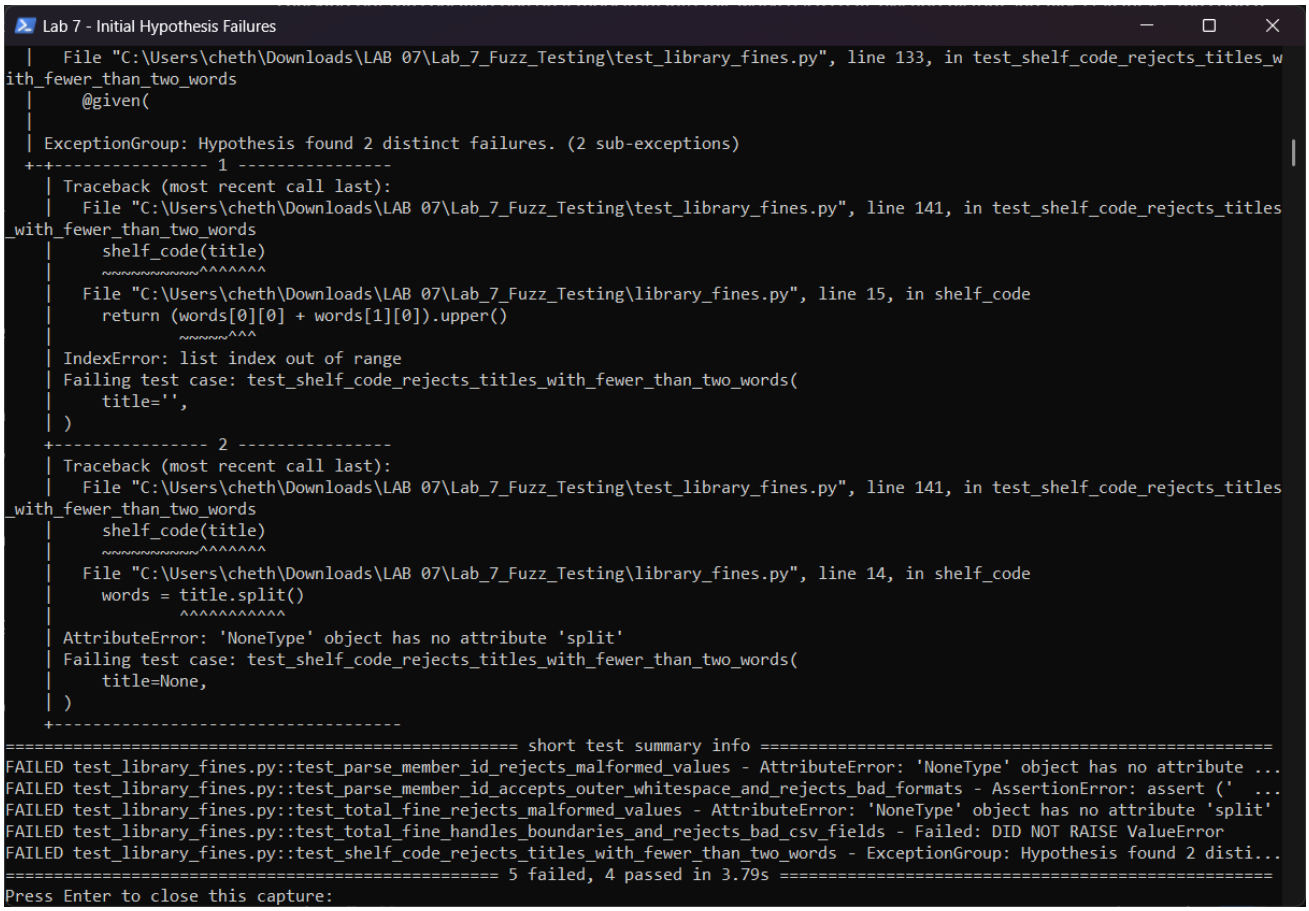
PES University | Student ID: 24AM426

1. Original application



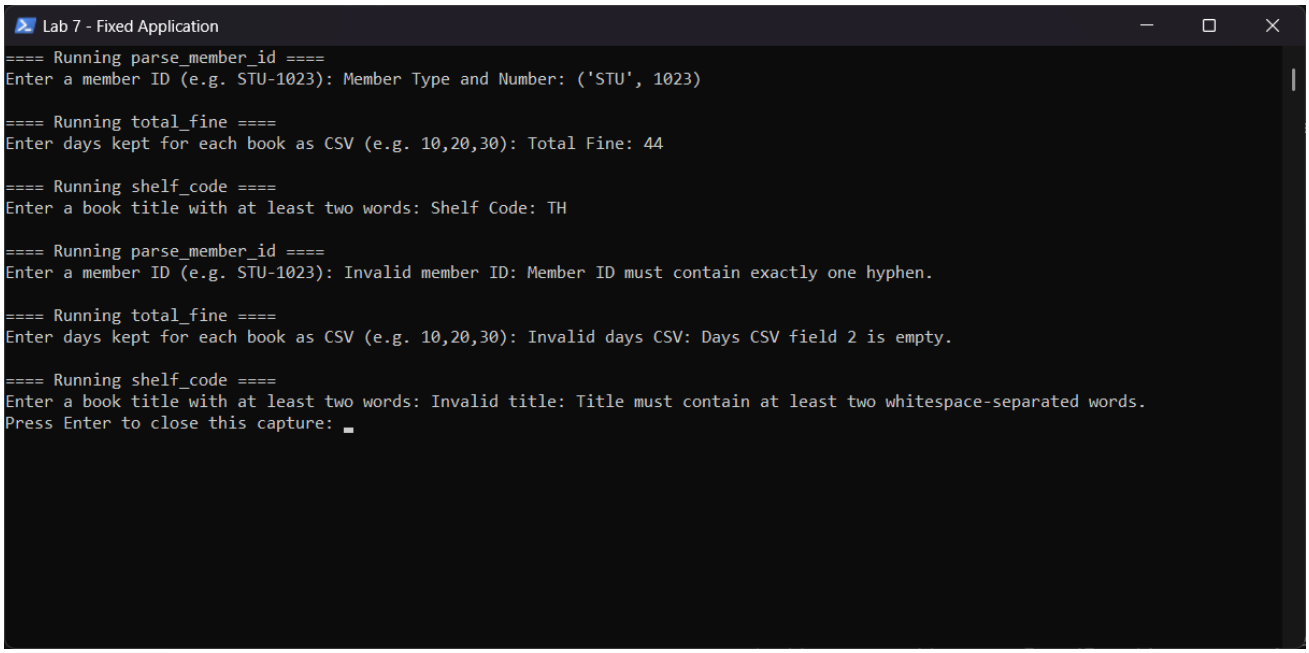
The original program completes a typical run with valid inputs. The following fuzz run checks edge cases that this happy-path execution does not reveal.

2. Initial Hypothesis failures



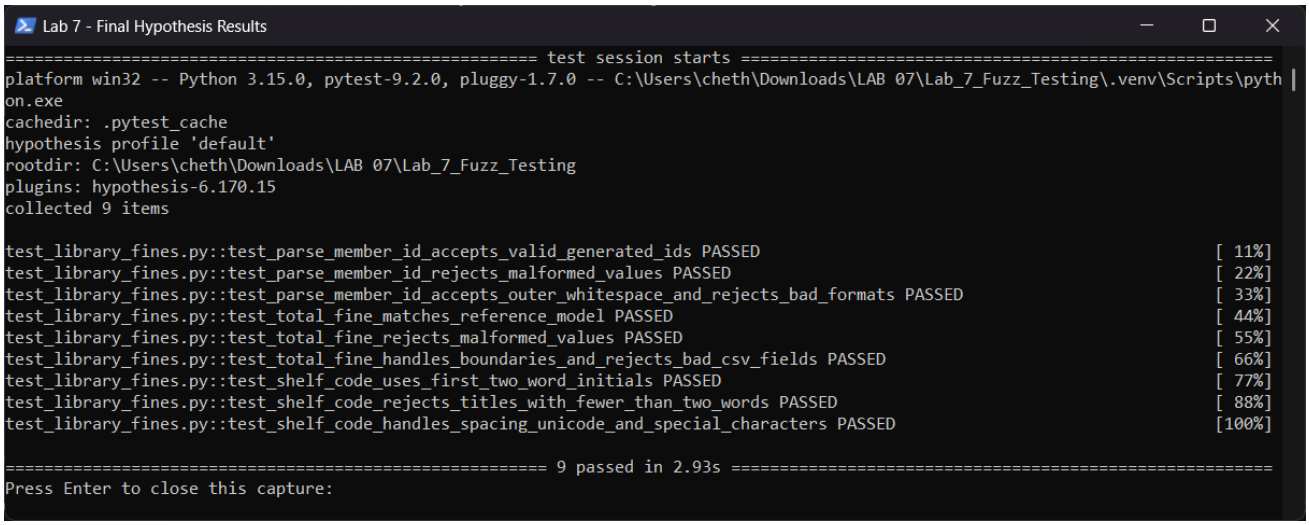
Hypothesis minimized unexpected inputs that triggered AttributeError and IndexError. The run also exposed whitespace handling and malformed CSV validation problems. Initial result: 5 failed and 4 passed.

3. Fixed application



The fixed program returns the expected results for valid input. In the same captured run, malformed IDs, CSV entries, and a one-word title produce clear validation messages, and the application continues.

4. Final Hypothesis results



All 9 pytest cases pass after the fixes. Six property tests each explore up to 120 generated examples across valid and malformed values for the three functions.